

CAMPUS CARBON

Universal Campus Carbon Footprint Assessment & Management Platform (v2.0)

Organization:	Christ University	Reporting Year:	2025
Campus Name:	Christ University ? Kengeri Campus	Assessment Status:	Completed
Campus Type:	University / College	Total Population:	6,450 occupants
Location:	Bengaluru, Karnataka, India	Built-up Area:	161,300.0 m ² (78.5 Acres)
Standard / Protocol:	GHG Protocol Corporate Standard / ISO 14064-1 (India CEA v19 Grid Baseline)	Generated On:	02 September 2026, 10:33

1. GROSS / POSITIVE EMISSIONS	2. ELIGIBLE REDUCTIONS & REMOVALS	3. NET CARBON FOOTPRINT
2,167.87	-132.82	2,035.05
tCO ₂ e	tCO ₂ e	tCO ₂ e

1. Executive Carbon Accounting Breakdown

Scope Classification	Activity Description	Emissions (tCO ₂ e)	Share of Gross (%)
Scope 1 (Direct)	Stationary DG diesel, canteen LPG, campus fleet combustion, refrigerants	375.95	17.3%
Scope 2 (Indirect - Electricity)	Purchased grid electricity & EV charging consumption	1,528.52	70.5%
Scope 3 (Other Indirect)	Commuting, solid waste disposal, freshwater supply, wastewater STP	263.40	12.2%
Total Gross Emissions	All combined operational activity emissions	2,167.87	100.0%
Eligible Removals / Reductions	Tree carbon sequestration, verified EV savings, waste diversion	-132.82	-6.1%
Official Net Carbon Footprint	Gross Emissions – Eligible Reductions & Removals	2,035.05	93.9%

Gross Intensity / Person:	Net Intensity / Person:	Gross Intensity / Area:	Renewable Energy Share:	Waste Diversion Rate:
0.3361 tCO ₂ e/person	0.3155 tCO ₂ e/person	13.44 kgCO ₂ e/m ²	32.8%	85.3%

2. Building-wise Carbon Performance Leaderboard

Code	Building Name	Type	Area (m ²)	Occ.	Gross (t)	Red. (t)	Net (t)	tCO ₂ e/p	kg/m ²
ENG-02	Engineering Block II (CSE...	Academic	16,200	1400	175.42	0.00	175.42	0.13	10.8
ENG-01	Engineering Block I (Civi...	Academic	14,500	1200	158.98	0.00	158.98	0.13	11.0
MGMT-01	School of Business & Mana...	Academic	11,500	900	139.75	0.00	139.75	0.16	12.2
HST-03	Carmel Hall Girls Hostel	Hostel	13,200	750	137.50	0.00	137.50	0.18	10.4
HST-01	St. Kuriakose Boys Hostel	Hostel	12,500	650	130.34	0.00	130.34	0.20	10.4
HST-04	Mother Teresa Girls Hoste...	Hostel	12,800	700	125.30	0.00	125.30	0.18	9.8
LIB-01	Central Knowledge & Libra...	Academic	7,400	500	114.56	0.00	114.56	0.23	15.5
HST-02	Devadan Hall Boys Hostel	Hostel	11,800	600	110.98	0.00	110.98	0.18	9.4
RES-01	Center for Research & Inn...	Academic	6,200	350	100.24	0.00	100.24	0.29	16.2
ARCH-01	School of Architecture & ...	Academic	9,800	600	82.34	0.00	82.34	0.14	8.4

QTR-01	Staff & Faculty Residenti...	Residential	8,200	200	78.76	0.00	78.76	0.39	9.6
BIO-01	Bio-Technology & Innovati...	Laboratory	9,200	450	75.62	7.92	67.70	0.17	8.2
BIO-02	Bio-Technology Research W...	Laboratory	9,200	450	75.62	7.92	67.70	0.17	8.2
ARTS-01	Social Sciences & Humanit...	Academic	8,500	700	68.02	0.00	68.02	0.10	8.0
AUD-01	Main Auditorium & Cultura...	Academic	5,800	800	58.74	0.00	58.74	0.07	10.1
TEST-01	Test Block	Academic	4,500	300	0.00	0.00	0.00	0.00	0.0

3. Negative / Eligible Reduction Contributions Ledger

Category	Activity & Intervention	Methodology & Baseline	Status	Avoided / Sequestered
Greenery	Biomass Sequestration by Campus Tre...	FSI Tree Biomass Growth Model & IPCC GPG...	Eligible	-61.60 tCO2e
EV Transition	EV Fleet Operation Savings vs Basel...	Baseline ICE Displacement (Fuel equivale...	Eligible	-0.00 tCO2e
Waste Diversion	Avoided Landfill Methane via Compos...	Avoided Landfill Methane Generation (IPC...	Eligible	-36.08 tCO2e
Waste Diversion	Avoided Landfill Methane via Recycl...	Avoided Landfill Methane Generation (IPC...	Eligible	-19.30 tCO2e
Waste Diversion	Avoided Landfill Methane via Compos...	Avoided Landfill Methane Generation (IPC...	Eligible	-7.92 tCO2e
Waste Diversion	Avoided Landfill Methane via Compos...	Avoided Landfill Methane Generation (IPC...	Eligible	-7.92 tCO2e

4. Prioritized Recommendations & Decarbonization Action Plan

Priority	Category	Intervention Action Title	Intervention Type	Est. Impact
High	Electricity	On-site Rooftop Solar PV Expansion & Sma...	Emission Reduction	-25%
High	Electricity	HVAC & Lighting Deep Energy Efficiency R...	Emission Reduction	-15%
Medium	Transport	Phased Fleet Electrification & EV Shuttl...	Emission Reduction	-20%
Medium	Transport	Commuter Carpooling & Green Transit Ince...	Emission Reduction	-10%
Medium	Greenery	Afforestation & Miyawaki Native Urban Fo...	Carbon Sequestration / Removal	-8%

5. Accounting Boundaries & Anti-Double-Counting Disclosures

Methodological Boundary: This assessment is executed under the operational control consolidation approach in strict accordance with the GHG Protocol Corporate Accounting and Reporting Standard and ISO 14064-1:2018 specifications.

Anti-Double-Counting Safeguards:

1. **On-site Solar Generation:** Solar energy consumed directly on campus displaces grid power and is reflected exclusively in reduced Scope 2 emissions. It is NOT subtracted a second time as a separate negative carbon offset.
2. **EV Fleet Accounting:** Operational EV charging electricity is quantified under Scope 2 electricity. Avoided transport reductions are calculated separately against an explicit ICE baseline.
3. **Carbon Sequestration:** Tree and vegetation biomass sequestration is modeled based on Forest Survey of India & IPCC GPG factors (22.0 kg CO2/tree/year) and reported in the eligible removals ledger.